7. Let V be a subset of the vector space $\boldsymbol{R^n}$ consisting only of the zero vector of $\boldsymbol{R^n}$, Namely $\boldsymbol{V} = \{\mathbf{0}\}$. Then prove that V is a subspace of $\boldsymbol{R^n}$.

Solution: To prove that $V = \{\mathbf{0}\}$ is a subspace of $\mathbb{R}^n$, we check the following subspace criteria.

Subspace Criteria

(a) The zero vector $\mathbf{0} \in \mathbb{R}^n$ is in V.

(b) If $\mathbf{x}, \mathbf{y} \in V$, then $\mathbf{x} + \mathbf{y} \in V$.

(c) If $\mathbf{x} \in V$ and $c \in \mathbb{R}$, then $c\mathbf{x} \in V$.

Condition (a) is clear since V consists of the zero vector $\mathbf{0}$.

To check condition (b), note that the only element in $V = \{\mathbf{0}\}$ is $\mathbf{0}$. Thus if $\mathbf{x}, \mathbf{y} \in V$, then both $\mathbf{x}, \mathbf{y}$ are $\mathbf{0}$. Hence

$$\mathbf{x} + \mathbf{y} = \mathbf{0} + \mathbf{0} = \mathbf{0} \in V$$

and condition (b) is met.

To confirm condition (c), let $\mathbf{x} \in V$ and $c \in \mathbb{R}$. Then $\mathbf{x} = \mathbf{0}$.

We have

$$c\mathbf{x} = c\mathbf{0} = \mathbf{0} \in V$$

and condition (c) is satisfied.

Hence we have checked all the subspace criteria, and hence the subset $V = \{\mathbf{0}\}$ consisting only of the zero vector is a subspace of $\mathbb{R}^n$.

What's the dimension of the zero-vector space?

What's the dimension of the subspace $\boldsymbol{V} = \{\mathbf{0}\}$?

The dimension of a subspace is the number of vectors in a basis. So let us first find a basis of V.

Note that a basis of V consists of vectors in V that are linearly independent spanning set. Since 0 is the only vector in V, the set $S = \{\mathbf{0}\}$ is the only possible set for a basis.

However, S is not a linearly independent set since, for example, we have a nontrivial linear combination $\mathbf{1} \cdot \mathbf{0} = \mathbf{0}$.

Therefore, the subspace $V = \{\mathbf{0}\}$ does not have a basis.

Hence the dimension of V is zero.